Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

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1 Introduction

The rise of Generative AI (GenAI) and its capabilities has become a major topic of discussion in the education industry [1]. Ever since the release of ChatGPT by OpenAI in November 2022, the education landscape has evolved tremendously [1]. Whereas institutions initially banned the use of AI, the trend has shifted to embracing and integrating GenAI into pedagogy [2]. This transition is supported by GenAI's ability to deliver personalized learning and customized feedback based on individual student needs [3]. By utilizing GenAI, students are now able to focus on higher-order skills such as critical thinking and creativity, while AI scaffolds and automates repetitive tasks [4, 5]. Moreover, due to today's rapidly evolving professional landscape, the adoption of GenAI in higher education has become a necessity rather than an option [6]. Hence, institutions need to create an environment where AI can be used to enhance human intelligence rather than replace it [1, 5].

However, the adoption of GenAI in higher education comes with a significant set of challenges. As previously mentioned, GenAI can be useful in taking over repetitive tasks [7], but this also leads to "cognitive offloading" where students outsource critical cognitive tasks to the AI [7]. The pattern of reliance starts from using AI as an information verification tool to slowly becoming a content delegation tool, causing students to blindly rely on AI to generate the final output [8]. It becomes even more concerning as AI creates a "productivity paradox", where despite students experiencing increased productivity, they lack depth in critical analysis [9]. For instance, due to convenience and speed that AI summaries provide, students tend to prioritize these summaries over verifying the underlying sources [10]. Despite the faster drafting process, the final result ended up being lower quality, with generic essays and a lack of critical analysis [1, 9]. Moreover, this matter is further supported by the so-called "death of essays", a phenomenon that has risen to prominence since COVID-19 due to the increase in take-home exams and students utilizing AI to cheat and plagiarize [1]. This raises the importance of AI literacy. Studies show that students with higher AI literacy tend to take a collaborative approach by planning and evaluating AI outputs, which consequently leads to better outcomes compared to students with lower AI literacy, who tend to over-rely on AI just to get the job done [11]. Additionally, over-reliance on AI can lead to a decline in face-to-face interactions, negatively impacting students' interpersonal skills and emotional intelligence. Thus, increasing the risk of social anxiety and isolation [12].

As previously mentioned, over-reliance on AI is a significant concern that must be addressed. AI tends to generate text with a convincing tone, giving an illusion that the output is always true, causing users to overlook hallucinations and in the end hallucinating themselves [1, 10]. To combat this, students are required to possess foundational knowledge before adopting the tool [1, 13]. This ensures they are able to critically validate the AI's information, as the human in the loop is responsible for the final output produced [1, 14]. Therefore, students must be taught to verify AI outputs [15]. On the other hand, educators are encouraged to create critical thinking assessments that can't be replicated by AI, such as video presentations or pitches [15, 1].

This gap creates a strong opportunity for designing how students should interact with AI. As previously mentioned, AI tutoring styles is not static [3]. The characteristics of the AI's interaction can be engineered using prompts (prompt engineering) and guardrails [16]. In other words, poor learning outcomes are not solely based on the AI's accuracy and performance, but also an issue with the interaction design itself [17, 18].

To address the illusion of truth, AI tools must be designed with guardrails that encourage cognitive engagement rather than handing out direct answers [19]. In the context of learning, the design of the AI interaction itself is more important than its capabilities [18]. Evidence shows that the use of scaffolding techniques, such as guiding hints and follow-up questions, are able to significantly improve students' learning outcomes [18]. AI chatbots with these guardrails are referred to as Socratic chatbots, which are designed to prevent giving out direct answers to encourage self-reflection and deep understanding of the topic [13, 20, 21]. When compared to default GenAI models, Socratic chatbots are able to produce 79% more learning improvements than the default models (127% vs. 48%) during practice assessment [18]. Additionally, unlike the default model where students experience learning loss when AI is removed, students did not experience any learning loss when the Socratic chatbot is removed [18]. This indicates that the default AI interaction may have short-term benefits, but confidence outpaces actual understanding, creating an illusion of competence [18]. Although the Socratic tutoring style works in many contexts, it is not universally superior [22]. In the context of coding, a study comparing the effects of Socratic AI versus normal AI shows that Socratic AI produces a higher error ratio and has no evidence of improving long-term outcomes [22]. Furthermore, learners with lower AI literacy tend to find scaffolding bots frustrating, which further amplifies the need for alternative tutoring styles [13]. This underscores the need to look beyond Socratic tutoring. It is important to explore other tutoring styles that could effectively improve both learning and confidence calibration.

Yet most literature rarely compares alternative interaction designs within the same study. Despite growing evidence on AI-supported learning, most studies evaluate one tutoring style at a time. As a result, the field still lacks clear comparative evidence on which tutoring style best results in balanced learning outcomes as well as psychological learning benefits. In practice, GenAI can be deployed with a spectrum of tutoring styles using prompt engineering

and guardrails [16]. Beyond Socratic approaches, evidence shows that reverse tutoring can be effective in boosting cognitive engagement and knowledge mastery [23, 24, 17, 25]. This pedagogical approach embodies the theory of learning-by-teaching by using recursive feedback that observes how a tutee applies what has been taught [23]. This approach is similar to designs where the AI takes the role of a student, prompting the human learner to teach the AI, thereby reinforcing the learner's own understanding [15]. Studies show that reverse tutoring creates the protégé effect where students learn harder when they know they must teach [23, 25]. There are two mechanisms that align with reverse tutoring [24]:

1. Generative processing: Students are encouraged to inherently organize and integrate their knowledge.
2. Metacognitive monitoring: Students are able to detect explanation gaps, which trigger self-correction.

Moreover, this socio-psychological interaction shifts the learners mindset to dig deeper into the topic, creating a more flexible mastery of the topic [24, 17].

Hence, in addition to the Socratic tutoring style, this study introduces the reverse tutoring technique. This study compares the three tutoring styles (direct-answer, Socratic, and reverse tutoring) within a single comparative framework, while assessing both cognitive and psychological outcomes. In particular, reverse tutoring, which is grounded in learning-by-teaching theory, despite showing clear effectiveness in improving learning outcomes, remains under-explored in the GenAI context. Therefore, this study aims to fill that gap by asking how different tutoring styles compare in their effects on learning performance, knowledge acquisition, self-efficacy, and confidence calibration. This combined evaluation is important as prior work shows that students can appear more productive with AI but could lose their learning once the tool is removed [9, 18]. Accordingly, this study aims to determine which tutoring style best supports durable learning rather than short-term productivity.

2 Literature Review

2.1 Generative AI in Education

It cannot be denied that AI has revolutionised the way the education industry operates [6]. In fact, AI has been implemented within the education industry for decades [21]. Introduced by Jaime R. Carbonell in 1970, the SCHOLAR tutor system was the first intelligent tutoring system (ITS) [26]. By utilizing a semantic network of concepts and facts, the system was able to evaluate students' knowledge of geography, creating a two-way interaction with the student [26]. Over the past decade, specialized AI tools such as Wolfram Alpha and Geogebra have enhanced STEM learning by increasing interactivity, while tools like Grammarly have assisted students in writing [21]. Although these tools were effective, they lack the conversational capabilities that current GenAI models possess. Driven by natural language processing (NLP), these large language models (LLMs) are able to hold sustained dialogue, and have since changed the way humans interact with information [21, 1].

The presence of GenAI has not only changed the way students and teachers interact, but also the entire curriculum at all levels of education [15, 2, 27]. From initially banning the use of AI due to concerns of cheating and plagiarism, institutions are now embracing and integrating AI into their pedagogy [2, 6]. This shift is supported by GenAI's ability to deliver an adaptive and personalized learning experience for students [3]. For a long time, education has been challenged in providing the unique needs of each individual student [21]. Evidence shows that students experience an increase in their learning outcomes when being tutored in a one-on-one setting [28]. However, the cost of providing human one-on-one tutoring is expensive in both financial and human resources, especially as teachers are also burdened with administrative responsibilities [28, 15]. This is where GenAI comes into play, by providing a scalable and cost-effective one-on-one tutor compared to traditional human tutors that cost $3,500 - -4$,300 per student per year [28]. Hence, lowering the entry barrier for complex subjects by providing examples and analogies that cater to the user's understanding and interest [2, 29]. For example, in the context of vocabulary learning, the AI's output can be personalized based on the user's hobbies such as cooking, sci-fi, or sports; turning a dry learning experience into a more engaging and interactive one [29, 1]. Additionally, AI aids overworked instructors by making complex pedagogy accessible [15]. Instructors are able to leverage AI to provide examples and analogies to address misconceptions and design materials that reinforce learning [15].

Despite the immense benefits that GenAI provides, when used without proper guidance, it can cause harm to long-term learning outcomes [18]. Therefore, it is critical for educators to standardize GenAI's use in education. In order to correctly adopt GenAI in their pedagogy, educators are encouraged to follow UNESCO's AI Competency Framework which acts as a guide for AI use and misuse [30]. The framework highlights several key dimensions essential to effective AI implementation: students must develop foundational knowledge, use the tool ethically, and ensure that a human remains responsible for overseeing and validating the final output [30, 14]. Thus, institutions must create an environment where AI enhances human intelligence rather than replace it, fostering higher-order thinking and creativity

[5, 1]. Moreover, GenAI should be prevented if it removes the opportunity for students to develop cognitive abilities or social skills [1].

However, in practice, GenAI is still being used incorrectly, leading to short-term productivity but long-term learning loss, known as the "productivity paradox" [9, 18]. This phenomenon occurs because students tend to offload cognitive tasks to the AI, such as planning, organising, and evaluating [7]. Evidence shows that most students prioritize the convenience and speed of AI summaries over verifying the underlying sources [10]. Hence, students who over-rely on AI are found to produce generic and lower-quality work [1]. Consequently, the excessive use of ChatGPT reduces their academic performance and memory [31]. This shows that GenAI cannot yet be supportive of self-regulated learning unless it is pedagogically engineered to provide hints and scaffold rather than direct answers [7]. To mitigate the misuse of GenAI, educators are encouraged to provide assessments that require students to document their process in reaching their final output (e.g. oral examination or AI-output documentation)[27, 1]. Furthermore, institutions are encouraged towards using GenAI to foster higher-order skills and creativity, combining traditional pedagogy with AI-assisted exploration [5, 2]. Thus, it is important to design a human-centric model where GenAI acts as a cognitive partner rather than a replacement [5].

2.1.1 Socratic AI versus Direct AI

As previously mentioned, without proper guardrails, GenAI outputs direct answers, imposing the risk of cognitive offloading [7, 5]. To prevent this, Socratic chatbots are designed by using prompt engineering and guardrails to encourage cognitive engagement by offering hints and guidance rather than direct answers [16, 20]. As the name suggests, the design of Socratic AI chatbots originated from the Socratic teaching method [21]. The method itself was originally designed to doubt all assumptions using questions and dialogues, forcing students to construct their own knowledge [32]. This technique proves to be effective in improving students' learning outcomes, improving their learning gain by 59% in the context of coding comprehension [33]. In addition, a comparative study comparing Socratic AI versus direct AI shows that Socratic AI is able to improve learning gain by 127% with no learning loss when the AI is removed [18]. In comparison, the default model shows a learning loss of 17% when the AI is removed [18]. The open-ended questions and reflective prompts are proven to be more useful in promoting self-regulated learning as agreed by 85% of educators [**?**]. This is due to the back-and-forth interaction that encourages students to verbalise their reasoning and correct any misconceptions in comparison to the default model that gives direct solutions that deprive students of cognitive effort [34]. Thus, it can be acknowledged that the design of the AI interaction is more important than the model capabilities in improving learning outcomes [18].

Additionally, the convincing tone of default AI models can lead to users believing every response is true, despite the inaccuracies caused by hallucinations [1, 35]. This "illusion of truth" can be harmful as it leads to users overlooking these hallucinations, causing errors in their final output [1, 10]. Subsequently, users' confidence tends to align with the AI-expressed confidence, further imposing the danger of this illusion [36]. Moreover, the authoritative tone of the AI leads to students being over-reliant on the AI, which is proven to limit creativity and critical thinking as well as social and emotional development [11, 36]. Furthermore, being over-reliant on the AI leads to a lack of face-to-face interaction which negatively impacts students' interpersonal skills that leads to a higher risk of social anxiety and isolation [12].

To mitigate this, Lo et al., proposed a CLEAR framework which stands for Critical, Reflective, Adaptive, Explainable, and Responsible [14]. This framework encourages students to critically analyse the AI's output, reflect on their own learning, adapt to the feedback, explain the reasoning behind their answer, and be responsible for the final output [14]. However, to be able to apply this framework, the student must possess foundational knowledge of the topic as well as AI literacy to critically evaluate the AI's output [14]. Simply flagging whether the content is fake is not effective as students must develop internal critical thinking to succeed [37]. A study comparing traditional study methods, studying with direct AI, and studying with Socratic AI approaches, found that users who study with direct AI showed a learning loss of 17% when the AI is removed, while users who study with the traditional method and Socratic AI showed no learning loss [18]. This further proves that Socratic AI chatbots go beyond fact-checking, as it allows students to recognize cognitive biases and build emotional triggers, encouraging them to become more cognitively active [37].

2.2 Learning strategies: direct, Socratic, and learning-by-teaching

2.3 Confidence calibration and psychological impacts

2.4 Research gap

3 Research Question

Research question: How do direct-answer, Socratic, and reverse-tutoring (protégé effect) generative AI tutoring styles compare in terms of (a) student learning performance and knowledge acquisition, (b) self-efficacy, and (c) confidence calibration?

Hypothesis: Reverse tutoring (student teaches the AI as a naive tutee) will lead to better knowledge mastery, stronger self-efficacy, and better alignment between confidence and actual competence than both the Socratic chatbot and the direct-answer chatbot, because teaching forces learners to organise knowledge and exposes gaps (protégé effect), reducing the illusion of competence while supporting deeper learning.

Optional sub-questions: Which style best supports long-term retention? Do self-efficacy and perceived difficulty differ across conditions?

4 Research Plan

4.1 Methodology

4.2 Data and experiment

4.3 Participants and ethics

4.4 Timeline

5 Conclusion

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